AQI Prediction

Using Logistic Regression

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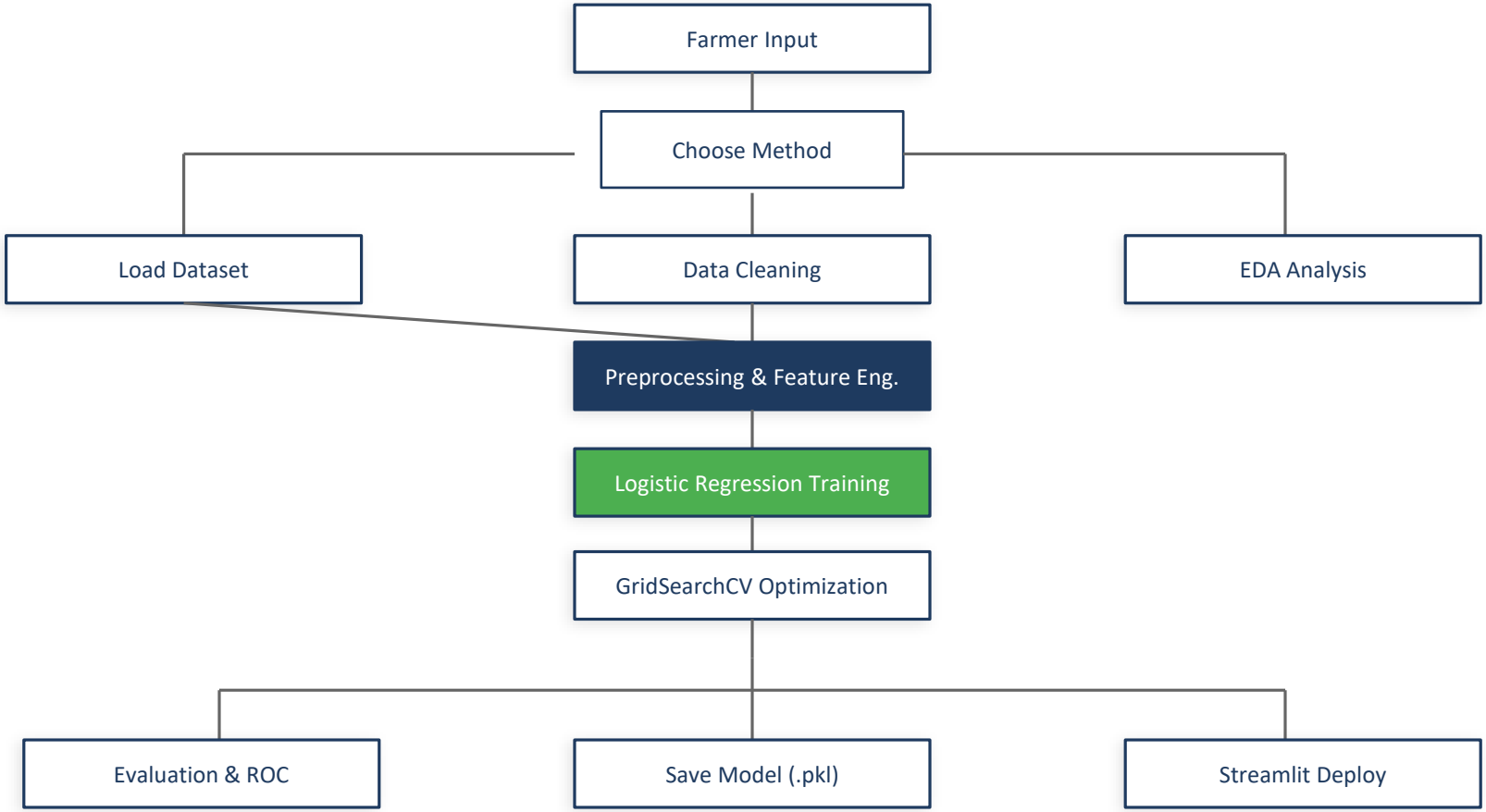
Problem Statement

1. Key Objectives

- Empower Farmers with AI: Provide Logistic Regression ML model to classify Air Quality Index levels from pollutant data.
- Real-Time Classification: Enable accurate prediction of AQI category (Good → Severe) using SO_2, NO_2, RSPM, SPM & PM2.5.
- Actionable Health Alerts: Deliver CPCB-standard category predictions with health advisories for public benefit.
- Sustainable Awareness: Help organizations and citizens understand pollution severity using data-driven insights.
- Multilingual Accessibility: Streamlit dashboard designed for a broad audience across technical levels.
- Knowledge Sharing: Open-source model artifacts and dataset pipeline for community reuse.

App Workflow Diagram



Brief Overview

Lack of Clean Air Data Insights: Most cities lack real-time AI-based classification of air quality, leading to delayed public health responses.

High Health Risk: Untreated exposure to Poor/Severe AQI zones causes respiratory disease, especially in children and the elderly.

Scientific Gap: Pollutant data from monitoring stations is raw and difficult for citizens or local bodies to interpret without expert knowledge.

Digital Divide: Modern pollution analytics are often English-only tools unavailable to regional government bodies and rural citizens.

Information Silos: No unified ML-based platform exists for predicting AQI categories from historical pollutant readings in real time.

Methodology

ML Pipeline Architecture: Built end-to-end using Python with Scikit-learn Logistic Regression (multinomial) as the core classification engine.

Multivariate Analysis Engine: Univariate, Bivariate, and Multivariate EDA using Matplotlib, Seaborn, and Plotly to understand pollutant distributions and correlations.

Disease Analysis Workflow:

Unified pollutant input via structured CSV (SO_2, NO_2, RSPM, SPM, PM2.5, Temperature, Humidity).

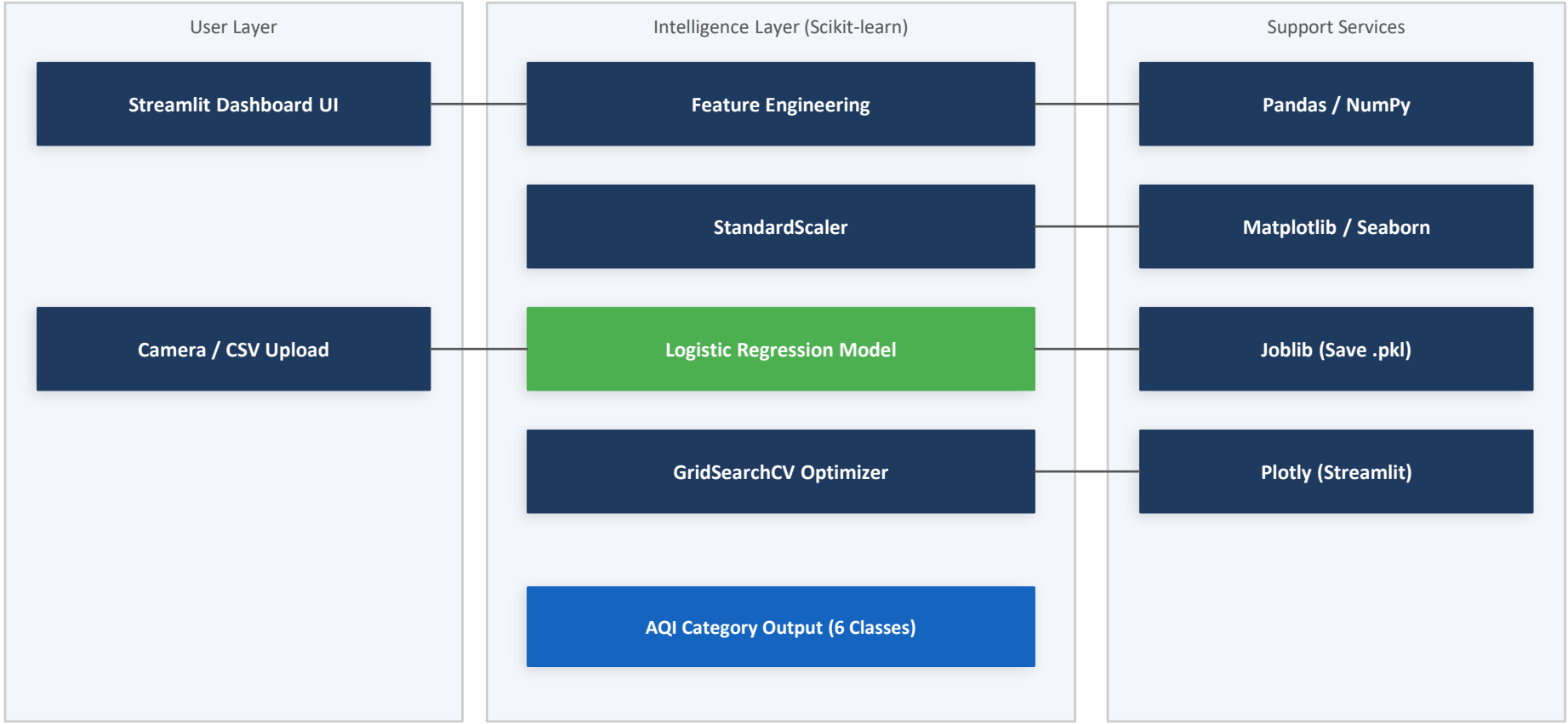
Automated feature engineering: Pollution Index, PM2.5/NO_2 ratio, Humidity-Temperature index.

Data Preprocessing: Label encoding for categorical variables, StandardScaler normalization, IQR-based outlier removal, city-level median imputation.

Model Optimization: GridSearchCV with 5-fold StratifiedKFold CV tuning regularization (C), solver, and max_iter.

Reporting: Model saved as .pkl via joblib. Streamlit app provides real-time prediction + health advisory per AQI class.

System Architecture



Algorithms Used

Logistic Regression (Multinomial): Core classification model using Softmax function: $P(Y=k|X) = \exp(\beta_k^T X) / \Sigma_j \exp(\beta_j^T X)$. Trained with L2 regularization via LBFGS/SAGA solver.

StandardScaler (Normalization): Z-score normalization applied to all features before training to ensure equal feature contribution and faster convergence.

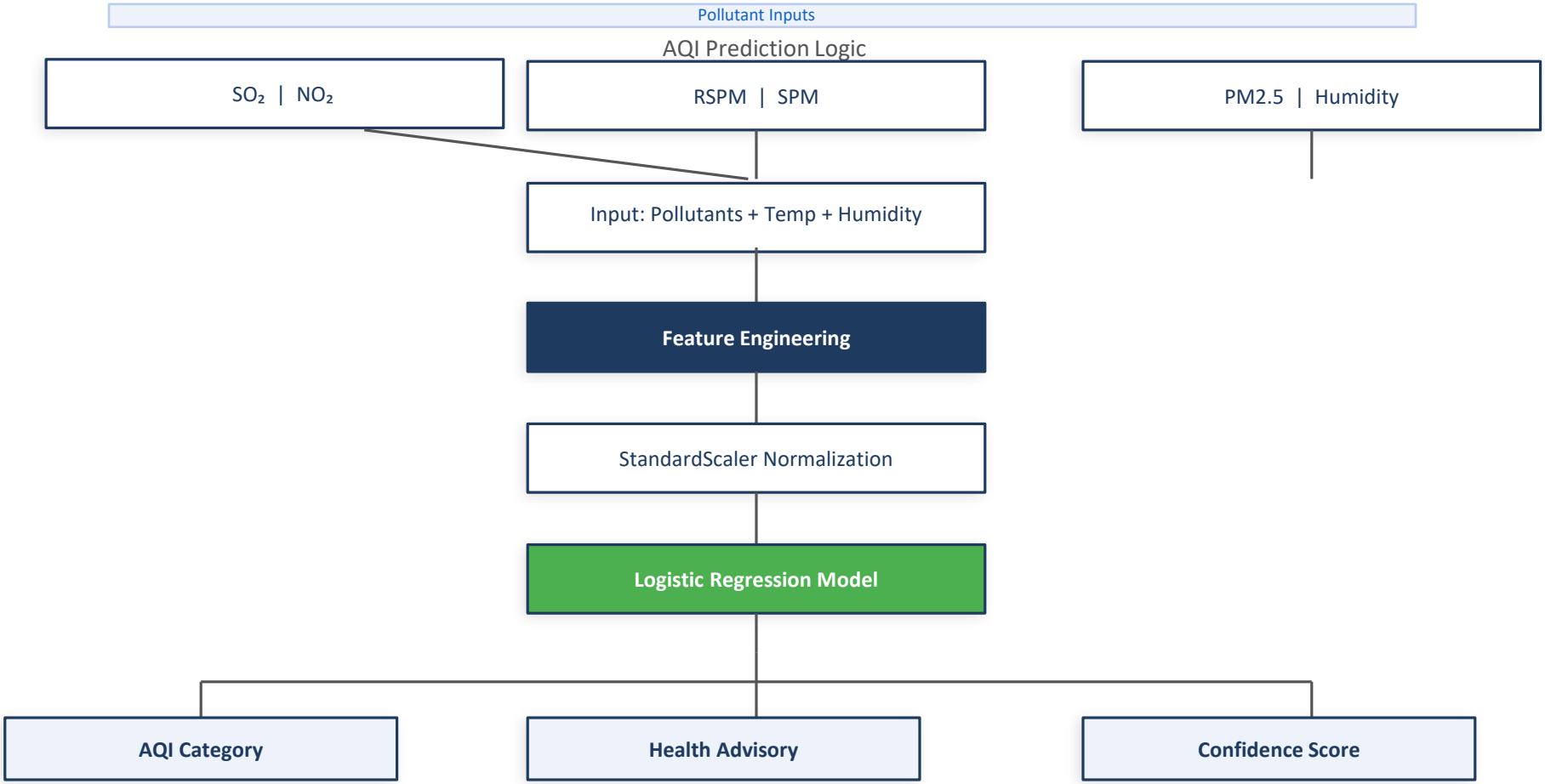
GridSearchCV (Hyperparameter Tuning): Exhaustive search over C (regularization), solver, and max_iter using 5-fold StratifiedKFold cross-validation.

Label Encoder: Encodes categorical variables (city, season) to numerical values. Target AQI categories encoded for multi-class classification.

IQR-based Outlier Removal: Interquartile Range (Q1 - 3×IQR, Q3 + 3×IQR) used to clip extreme pollutant readings from the dataset.

Feature Engineering: Pollution Index = $0.4 \times PM2.5 + 0.25 \times NO_2 + 0.2 \times RSPM + 0.15 \times SO_2$. $PM2.5/NO_2$ ratio and humidity-temperature interaction features added.

Methodology



Result and Output

Visual Health Metrics: Real-time AQI category prediction (6 CPCB classes: Good → Severe) with confidence probability per class.

Precision Classification:

STOP: Severe/Very Poor AQI — Immediate outdoor restriction advisory.

CAUTION: Moderate/Poor AQI — Preventive health measures recommended.

GO: Good/Satisfactory AQI — Outdoor activities safe to proceed.

Smart Recommendations: City and season-specific AQI predictions tailored to input pollutant readings.

Tangible Assets: Downloadable insights via Streamlit UI. Model saved as .pkl file for reuse.

88.7%

Accuracy

0.879

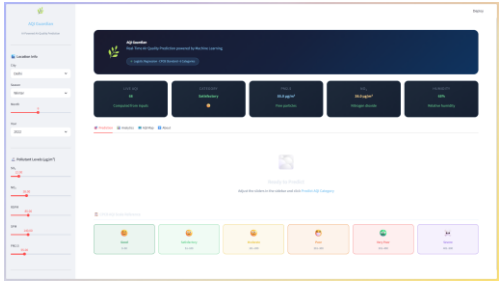
F1 Score

0.937

ROC-AUC

Optimized Model Performance

Prediction Dashboard

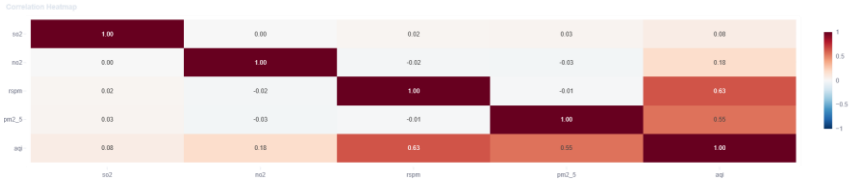


AQI Gauge + Class Probabilities

Analytics Tab

City AQI Comparison Charts

EDA Heatmap



Pollutant Correlation Matrix

India AQI Map

Plotly Mapbox City Bubbles

Requirements — Libraries & Dependencies

Core ML & Data

- ✓ pandas >= 1.5
- ✓ numpy >= 1.23
- ✓ scikit-learn >= 1.2
- ✓ joblib >= 1.2

Visualization

- ✓ matplotlib >= 3.7
- ✓ seaborn >= 0.12
- ✓ plotly >= 5.14

Deployment (UI)

- ✓ streamlit >= 1.28
- ✓ streamlit-option-menu

Notebook & Utils

- ✓ jupyter / google colab
- ✓ ipykernel
- ✓ warnings (built-in)
- ✓ os (built-in)

```
$  pip install pandas numpy scikit-learn matplotlib seaborn plotly streamlit joblib
```

Conclusion

- AQI Prediction ML model bridges the gap between raw pollutant data and actionable public health decisions by classifying air quality into 6 CPCB categories with 88.7% accuracy.

- By combining PM2.5, NO_2, RSPM, SPM, SO_2 with weather features, the model captures real-world pollution dynamics and achieves a mean ROC-AUC of 0.937 across all classes.

- The Streamlit deployment provides an interactive, city-centric dashboard with real-time AQI gauges, pollutant contribution charts, India AQI maps, and CPCB-standard health advisories.

- Scalability: The architecture is designed to integrate with live CPCB API sensor feeds, IoT pollution sensors, and satellite imagery (Sentinel-5P) for higher precision in the future.

Future Scope

Advanced ML Models: Replace Logistic Regression with XGBoost, Random Forest, or Neural Networks. Ensemble stacking for higher accuracy across all AQI categories.

Real-Time API Integration: Connect to live CPCB / OpenAQ pollution sensor APIs and OpenWeatherMap for true real-time predictions instead of historical CSV inputs.

Time-Series Forecasting: Add LSTM-based 7-day AQI forecast so citizens and authorities can plan outdoor activities and emergency protocols in advance.

National Scale Deployment: Scale to all 730+ Indian districts. Develop a mobile app + WhatsApp bot for last-mile health alert delivery in regional languages.

Policy Dashboard: Provide state governments with trend analytics, pollution source identification, and policy simulation to meet NAAQS standards.

Global Expansion: Extend model globally using Sentinel-5P satellite data to support WHO air quality monitoring in developing countries worldwide.

Thank You

AQI Prediction Project — GitHub Link:

https://github.com/Quickflashtech/AQI_Prediction_project/

QR CODE OF GITHUB